



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
12-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TNAS)

Time : 3 Hours

Maximum Marks : 216

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

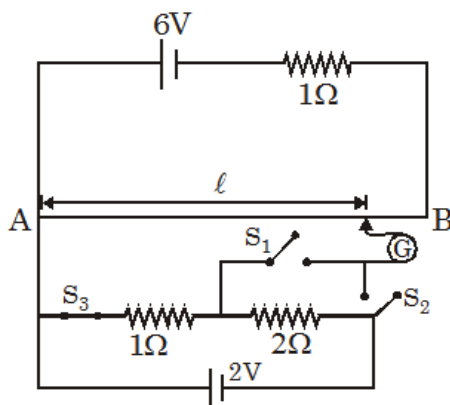
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

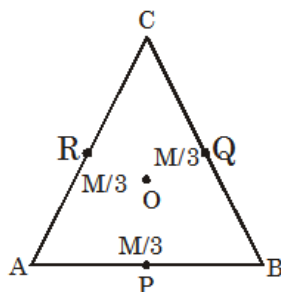
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Figure shows a potentiometer circuit. The length of potentiometer wire AB is 120 cm and its resistance is 2Ω . The internal resistance of 2V cell is 1Ω . Initially the switches S_1 and S_2 are open and S_3 is closed.



- (A) When only S_2 is open, balanced length $\ell = 20$ cm.
 (B) When only S_1 is open, balanced length $\ell = 60$ cm
 (C) When all three switches are closed, balanced length is 60 cm.
 (D) When S_3 is open, S_2 is also open but S_1 is closed, balance length is 60 cm.

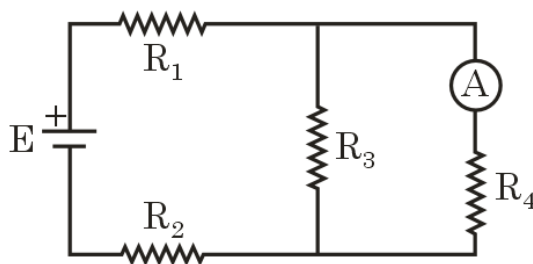
2. A thin rod of length ℓ is kept vertically (along y-axis) with end A being at rest on a smooth horizontal plane. A slight disturbance makes the end A slip on the plane (along x-axis) and the rod falls. Then,
- (A) the path described by a point P at a distance of $\frac{\ell}{4}$ from A is a quadrant of a circle.
- (B) the centre of mass of the rod moves along a straight line, parallel to y-axis.
- (C) the acceleration of the centre of mass as the rod becomes horizontal is $\frac{3g}{4}$.
- (D) the greatest angular velocity of the rod about the instantaneous centre of rotation is $\sqrt{\frac{3g}{\ell}}$.
3. A, B and C are the vertices of a uniform equilateral triangular thin lamina of mass M. Three particle of masses $\frac{M}{3}$, each are located at the midpoints P, Q and R of the edges of the lamina. For this system, (each edge of lamina = a)



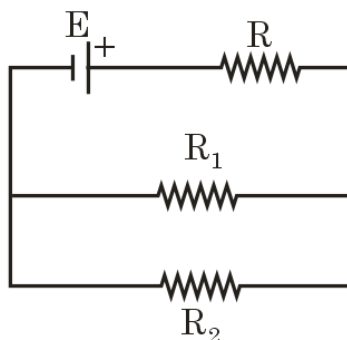
- (A) the moment of inertia of the system about an axis through the centroid O perpendicular to the plane is $\frac{Ma^2}{6}$.
- (B) the moment of inertia of the system about an axis through O perpendicular to the plane is $\frac{5Ma^2}{12}$.
- (C) the moment of inertia of the system about an axis through A perpendicular to the plane is $\frac{13Ma^2}{12}$.
- (D) the moment of inertia of the system about an axis through C perpendicular to the plane is $\frac{5Ma^2}{6}$.

Space for Rough Work

4. A network of four unequal resistors R_1 , R_2 , R_3 and R_4 are connected in a network (see diagram) that contains a source of e.m.f. E and an ammeter A . The source has negligible internal resistance. For this network.



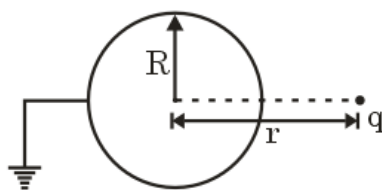
- (A) If $R_3 > R_4$, the current in the resistor R_3 will be less than that shown by the ammeter.
- (B) If $R_3 < R_4$, the current in the resistor R_2 will be greater than that shown by the ammeter.
- (C) If the source of e.m.f. and the ammeter were physically interchanged in the network the ammeter will show the same current.
- (D) If the resistors R_2 and R_1 were physically interchanged, the ammeter will show the same current
5. In the given network, the powers generated in the resistors R_1 and R_2 are P_1 and P_2 and the power generated in R is P . For this network,



- (A) If $P_1 = P_2$, it necessarily follows that $R_1 = R_2$
- (B) If $R = R_1 + R_2$, then $P > P_1 + P_2$
- (C) If $P = P_1 + P_2$, then $R \leq \frac{R_1 + R_2}{4}$
- (D) If $R_1 = R_2$, then it necessarily follows that $P = P_1 + P_2$

Space for Rough Work

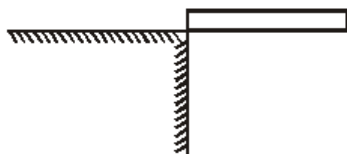
6. A charge q is placed at a distance r from centre of conducting sphere as shown in figure. Now sphere is earthed then correct options are :-



- (A) Induced charge on sphere is $-\left(\frac{R}{r}\right)q$
- (B) Induced charge on sphere is $-q$
- (C) Force acting between sphere and external charge q is $\frac{1}{4\pi\epsilon_0} \frac{q^2 R r}{(r^2 - R^2)^2}$
- (D) Force between sphere and external charge q is $\frac{q^2}{r^2}$.
7. A uniform disc of mass 2 kg and radius 1 m is mounted on a horizontal axle supported on fixed frictionless bearings. A light chord is wrapped around the rim of the disc and mass of 1 kg is tied to the free end. If it is released from rest (take $g = 10\text{ m/s}^2$)
- (A) the tension in the chord is 5 N .
- (B) in first four seconds the angular displacement of the disc is 40 rad .
- (C) the work done by the torque on the disc in first four seconds is 200 J .
- (D) the increase in the kinetic energy of the mass in the first four seconds is 200 J .

Space for Rough Work

8. A uniform rod of length ℓ is falling down with a velocity v_0 when one of its end hits a fixed edge as shown. (Consider impulsive force to be acting perpendicular to the length of rod)



- (A) If the collision is elastic, the centre of mass may have upwards velocity just after the collision.
- (B) If the collision is inelastic, the centre of mass may come to rest just after the collision.
- (C) If the collision is perfectly inelastic, the centre of mass has a downward velocity just after the collision.
- (D) If the collision is elastic, the centre of mass has a downward velocity just after the collision.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 9 and 10

A small particle of mass m is fixed to the perimeter of a ring of same mass and radius ' r ' and ring is placed on a horizontal plane. Friction is negligible on horizontal plane. Initially particle is at top most point, then the system is released from rest. Answer next two questions when the particle is at the same height as the centre of ring after being released from topmost point. Assume that the ring stays in vertical plane during its motion under consideration.



Space for Rough Work

9. Mark the correct option(s) :-

(A) Speed of centre of mass of system is $\sqrt{\frac{gr}{4}}$.

(B) Angular velocity of system is $\omega = \sqrt{\frac{g}{r}}$

(C) Contact point of ring with ground is instantaneously at rest.

(D) Centre of ring is instantaneously at rest.

10. Mark the correct option(s) :

(A) Instantaneous acceleration of the centre of ring is $\frac{g}{2}$.

(B) Acceleration of centre of ring initially points towards left and then towards right.

(C) The force exerted by the ring on the ground at the instant given is $\frac{3}{2}mg$.

(D) The force exerted by the ring on the ground at the instant given is $2mg$.

Space for Rough Work

Paragraph for Questions 11 and 12

A rocket is launched from and returns to a spherical planet of radius R in such a way that its velocity vector on return is parallel to its launch vector. The angular separation at the centre of the planet between the launch and arrival points is θ .

11. Which of the following can be the distance of the rocket above the surface of the planet ?
- (A) R (B) $\frac{R}{2}$ (C) $\frac{R}{3}$ (D) $4R$
12. How long does the flight of rocket take, if the period of satellite flying around the planet just above its surface is T_0 ?
- (A) $T_0 \left(\frac{1}{2} + \frac{1}{\pi} \right)$ if $\theta \rightarrow 0$ (B) $T_0 \left(\frac{1}{2} + \frac{1}{\sqrt{2}\pi} \right)$ if $\theta = 90^\circ$
- (C) $T_0 \left(\frac{1}{2} + \frac{2}{\pi} \right)$ if $\theta \rightarrow 0$ (D) $T_0 \left(\frac{1}{2} + \frac{\sqrt{2}}{\pi} \right)$ if $\theta = 90^\circ$

Space for Rough Work

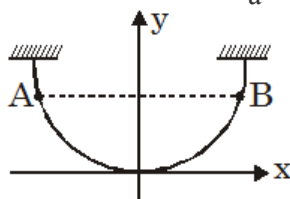
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

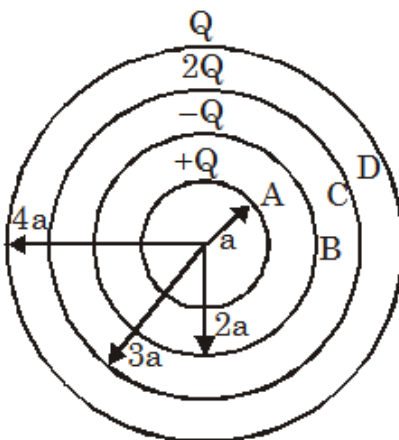
Zero Marks : 0 In all other cases.

1. A particle is uncharged and is thrown vertically upward from ground level with a speed of $5\sqrt{5}$ m/s. As a result, it attains a maximum height h . The particle is then given a positive charge $+q$ and reaches the same maximum height h when thrown vertically upward with a speed of 13 m/s. Finally, the particle is given a negative charge $-q$. Ignoring air resistance, determine the speed (in m/s) with which the negatively charged particle must be thrown vertically upward, so that it attains exactly the same maximum height h .
2. A smooth parabolic wire track defined by relation $y = \frac{x^2}{2a}$ is fixed in x-y plane in a gravity free space as shown in figure. An electric field $\vec{E} = \alpha y\hat{i} + \alpha x\hat{j}$ (where α is a constant) exist in the region. A small ring the mass ' m ', charge ' q ', which just fits in wire track, is released from rest at position A $\left(-a, \frac{a}{2}\right)$. Find speed of the ring when it reaches at point B $\left(a, \frac{a}{2}\right)$ if $q\alpha = \frac{2m}{a^2}$.



Space for Rough Work

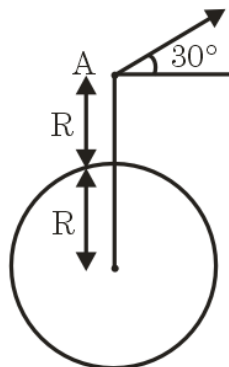
3. Consider four concentric conducting spherical shells of radius a , $2a$, $3a$ and $4a$ having charges Q , $-Q$, $2Q$ and Q respectively as shown in figure. Total electrostatic potential energy of this charge configuration is $\left(\frac{nQ^2}{\pi\epsilon_0 a}\right)$. Find n .



4. A voltmeter coil has resistance $50.0\ \Omega$ and a resistance of $1.15\ \text{k}\Omega$ is connected in series. It can read potential differences up to 12 volts. If this same coil is used to construct an ammeter which can measure currents upto $2.0\ \text{A}$, what should be the resistance of the shunt used ?

Space for Rough Work

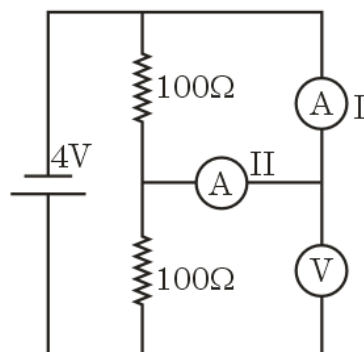
5. A particle of mass m is fired from the point A (as shown in the figure) situated at a distance of $2R$ from the centre of the planet with a velocity $\frac{1}{\sqrt{2}}$ times the escape velocity from point A. The radius of the planet is R . The maximum distance of the particle from the centre in the subsequent motion is nR . Find the value of n .



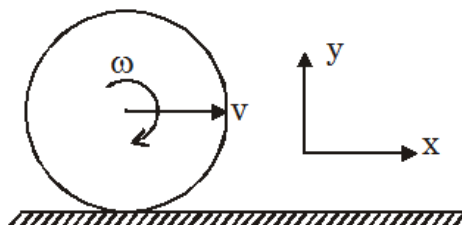
6. A metal sphere of radius a is surrounded by a concentric metal spherical shell of radius b ($b > a$). The space between the spheres is filled with material whose electrical conductivity σ varies with the electric field strength E according to the relation $\sigma = KE$ where K is constant. A potential difference V is maintained between inner sphere and outer shell. If the current (in ampere) between the inner sphere and outer shell is x , then find $x/4$. [Here $K = \frac{1}{\pi} \times 10^{-2} \text{ ohm}^{-1} \text{ volt}^{-1}$, $V = 20$ volts, $b = ea$ ($e = \text{exponential}$) and $\pi = 22/7$]

Space for Rough Work

7. In the figure ammeter (I) reads a current of 10 mA, while the voltmeter reads a potential difference of 3V. What does ammeter (II) (in mA) read ? The ammeters are identical, the internal resistance of the battery is negligible. (Consider all ammeters and voltmeters as non ideal]



8. A disc of radius '5cm' rolls on a horizontal surface with linear velocity $\mathbf{v} = 1\hat{i}$ m/s and angular velocity 50 rad/sec. Height of particle from ground on rim of disc which has velocity in vertical direction is (in cm) :



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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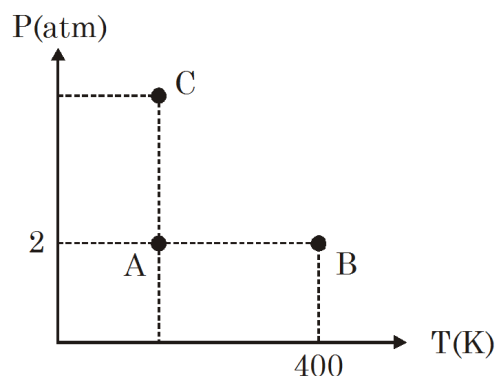
Negative Marks : -2 In all other cases.

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-
1. Three complex compounds I, II and III are $[M(H_2O)_6]^{3+}$, $[M(en)_3]^{3+}$, $[M(CN)_6]^{3-}$ respectively. Complex I and III have green and red colour respectively then identify the correct possible matching colour for complex II.
 (A) Blue (B) Orange (C) Yellow (D) Violet
 2. Which of the following hydrolysis process is partial hydrolysis
 (A) $XeF_6 + 3H_2O \rightarrow$ (B) $SbCl_3 + \text{excess } H_2O \rightarrow$
 (C) $BF_3 + \text{excess } H_2O \rightarrow$ (D) $PCl_5 + H_2O (1\text{mol}) \rightarrow$
-

Space for Rough Work

3. Which of the following solubility order will not be same in both water and CCl_4 medium?
- (A) $\text{AgCl} > \text{AgBr} > \text{AgI}$ (B) $\text{LiBr} > \text{NaBr} > \text{KBr}$
- (C) $\text{PbCl}_2 > \text{PbBr}_2 > \text{PbI}_2$ (D) All of these
4. $\frac{V_{\text{final}}}{V_{\text{initial}}} = 2$ is achieved for each of the three different reversible processes as mentioned below. (1 mole of ideal gas with $\gamma = 1.5$ for all processes)
- (x) Isothermal process ending at state - A
- (y) Isobaric process starting from state - A
- (z) Reversible adiabatic process ending at state - B



Which of the following statements is/are correct ?

- (A) Temperature of gas increases in two of the processes.
- (B) Maximum temperature observed during any of the processes is 400 K.
- (C) If "C" is initial state of isothermal process then $P_C = 4$ atm.
- (D) Initial pressure of adiabatic process is more than initial pressure of isothermal process if "C" is initial state of isothermal process.

Space for Rough Work

5. For the gas phase exothermic reaction, $A_2 + B_2 \rightleftharpoons C_2$, carried out in a closed vessel, the equilibrium moles of A_2 can be increased by

(A) increasing the temperature (B) decreasing the pressure
(C) adding inert gas at constant pressure (D) removing some C_2

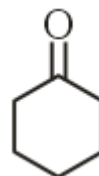
6. $(P) + NaHSO_3 \rightleftharpoons (Q)$

Compound (P) having molecular formula ' C_3H_6O ' & gives (+ve) haloform test. Choose the correct statement(s)?

(A) (Q) is $\begin{array}{c} H_3C \quad \quad OH \\ \quad \diagdown \quad \diagup \\ \quad C \\ \quad \diagup \quad \diagdown \\ H_3C \quad \quad SO_3Na \end{array}$ and crystalline in nature

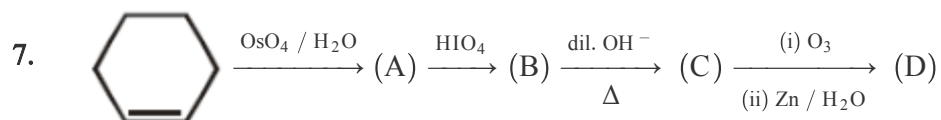
(B) (Q) is $\begin{array}{c} H_3C \quad \quad OH \\ \quad \diagdown \quad \diagup \\ \quad C \\ \quad \diagup \quad \diagdown \\ H_3C \quad \quad SO_3Na \end{array}$ and non-crystalline in nature

(C) (P) can be aldehyde, methyl ketone & some cyclic ketones as



(D) (Q) is $\begin{array}{c} H_3C \quad \quad OH \\ \quad \diagdown \quad \diagup \\ \quad C \\ \quad \diagup \quad \diagdown \\ H_3C \quad \quad OSO_2Na \end{array}$ and crystalline in nature

Space for Rough Work



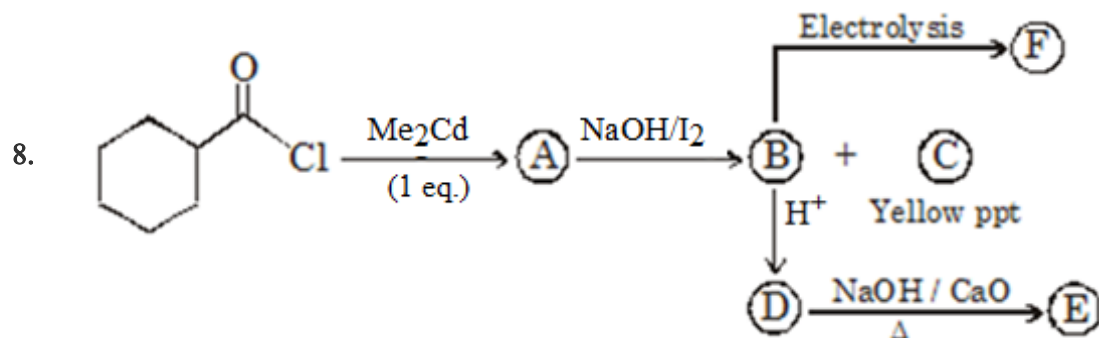
Correct statement(s) is/are

(A) (A) is vicinol diol

(B) (B) is a dicarboxylic acid

(C) (B) gives Tollen's test

(D) (D) gives Iodoform test



Correct statement is/are :

(A) Compound F is 

(B) Compound E is 

(C) A is acetophenone

(D) In reaction from A to B, 3 equivalent of I_2 consumed, with respect to 1 equivalent of A.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
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Paragraph for Questions 9 and 10

An octahedral homoleptic high spin complex of '3d' series central metal ion having $3d^n$ electronic configuration have $t_{2g}^x e_g^2$ arrangement of electrons.

Answer the following based on above paragraph.

9. Choose the correct pair of central atom/ion and ligand for the above mentioned complex.
- (A) Fe^{2+} , NH_3 (B) Co^{3+} , NH_3 (C) Fe^{3+} , F^- (D) Ni^{2+} , H_2O
10. For the maximum possible value of 'x' the value of spin only magnetic moment will be.
- (A) $\sqrt{8}$ B. M. (B) $\sqrt{15}$ B. M. (C) $\sqrt{24}$ B. M. (D) $\sqrt{35}$ B. M.

Space for Rough Work

Paragraph for Questions 11 and 12

A 10% by mass of sugar has undergone partial inversion according to the reaction



After 5 hours from the start of the reaction the elevation in boiling point of solution is $(5/18)^{\circ}\text{C}$. The reaction follows first order kinetics with respect to sugar and K_b for water is $0.57 \text{ kg K mol}^{-1}$.

Molar rotation of sugar, glucose and fructose are 60° , 50° and -90° respectively then.

11. Which of following options are correct :

(A) Half life of reaction is 5 hrs

(B) After $\left[5 \frac{\ln(5/2)}{\ln 2}\right]$ hours solution will be optically inactive

(C) After 10 hours solution will be laevo rotatory

(D) After 4 hours solution will be dextro rotatory

12. After completion of reaction solution is heating to 105.7°C temperature. If boiling point of pure water is 100°C then incorrect options will be :

(A) 82.15% mass of initial mass of water will be in vapour form

(B) Almost 93.5% mass of initial mass of water will be in vapour form

(C) Almost 5.84% mass of initial mass of water will be in liquid form

(D) Almost 93.5% mass of initial mass of water will be in liquid form.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

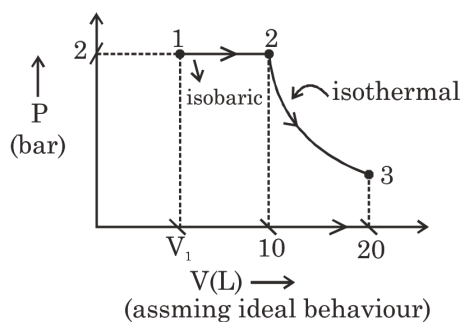
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Considering geometrical isomers of an octahedral complex $\text{Ma}_2\text{b}_p\text{C}_q$ have 'X' number of cis isomer w.r.t. 'a' ligand and 'Y' number of trans isomers w.r.t. 'a' ligand. Find the value of (X+Y)
(Consider the maximum possible value of X and Y)
a, b, c are monodentate achiral ligands while p and q are non-zero positive values.
2. If all the bond angles w.r.t. 'N' and 'B' are assumed as 120° in $\text{BH}_2(\text{NMe}_2)$ molecule then maximum number of atoms in one plane will be :-
3. A solid cube with edge length of 2.532 cm of an ionic compound which has NaCl type lattice is added to 1kg of water. The boiling point of this solution is found to be 100.52°C (Assume 100% ionisation of ionic compound). If radius of anion of ionic solid is $2\text{\AA}$, then calculate radius of cation in $\text{\AA}$.
Given : K_b of water = $0.52 \text{ K}\cdot\text{kg mol}^{-1}$,
 $N_A = 6 \times 10^{23}$, $\sqrt[3]{216} = 6$, $(2.532)^3 \times \frac{4}{3} = 21.6$

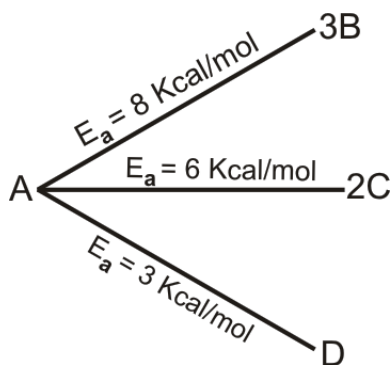
Space for Rough Work

4.



CO₂ gas in a conducting container fitted with a massless and frictionless piston is expanded from V₁ to 20 L as shown in the graph. The gas absorbs 2 kJ of heat during complete process from 1 to 3 then find the volume V₁ in litres. Assume 50% of vibrational degrees of freedom of CO₂(g) to be active and ideal behaviour (Use $\ln 2 = 0.7$) (1 litre - bar = 100 J)

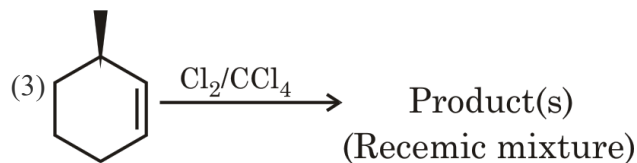
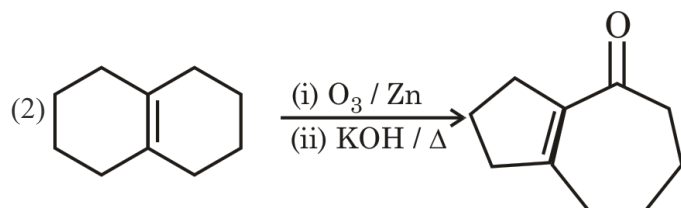
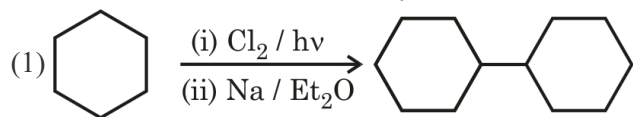
5. A substance undergoes first order parallel reaction as shown. Calculate net activation energy (in kcal/mol) if it is known that B, C and D are formed in a molar ratio of 2 : 3 : 4 respectively. Express answer in kcal.



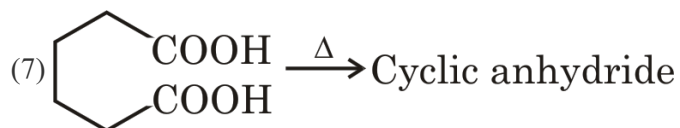
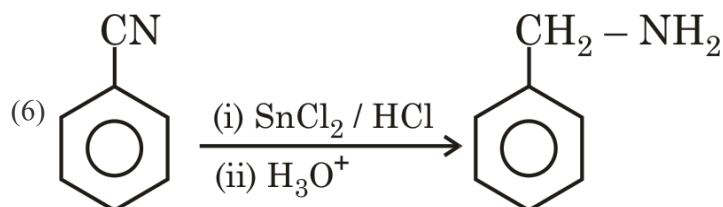
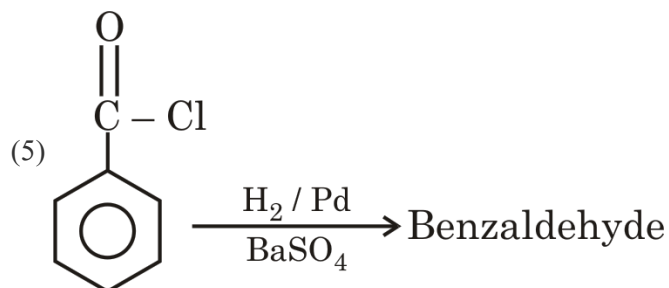
6. R-2-chlorohexane $\xrightarrow[h\nu]{\text{Cl}_2}$ Number of monochloro products.

Space for Rough Work

7. Find number of reaction correctly matched, with their major product (or statement written with them) :



(4) CH_3CHO gives positive Tollen's test as well as Lucas test



8. Which of the following compound when reacts with CH_3MgCl gives methane gas
(i) Tertiary butylchloride (ii) Tertiary butylalcohol (iii) Di-butylether (iv) Phenol
(v) Pyrole (vi) Pyridine

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. The value of $\lim_{x \rightarrow \infty} x^2 \tan \left(\ln \left(\sqrt{\cos \frac{\pi}{x}} \right) \right)$ is equal to

- (A) $\frac{-\pi^2}{2}$ (B) $\frac{\pi^2}{2}$ (C) $\frac{-\pi^2}{4}$ (D) $\frac{\pi^2}{4}$

2. If $\int \frac{x^3 - 8}{x^3 \sqrt{x^4 - x^3 + 2}} dx = \frac{k \cdot \sqrt{x^4 - x^3 + 2}}{x^2} + C$, then the value of k is - (where 'C' is integration constant)

- (A) 3 (B) 2 (C) 7 (D) 4

Space for Rough Work

3. If $\int \frac{3z^3 - 8z + 5}{\sqrt{z^2 - 4z - 7}} dz = (z^2 + az + 36) \sqrt{z^2 - 4z - 7} + b \ln |z - 2 + \sqrt{z^2 - 4z - 7}| + C$, where $a, b \in \mathbb{I}$ and C is integration constant, then -
- (A) $a > b$
 (B) $a < b$
 (C) $a + b = 117$
 (D) exactly one out of a or b is a prime number
4. Let $f(x) = \cos^{-1} \left(\ln \left(\frac{x+1}{x} \right) \right)$, then which of the following statement(s) is/are correct
- (A) Domain of $f(x)$ contains $(-\infty, -1) \cup (1, \infty)$
 (B) Range of $f(x)$ contains $\left[0, \frac{1}{e-1}\right]$
 (C) Domain of $f(x)$ is $\left(-\infty, \frac{e}{1-e}\right] \cup \left[\frac{1}{e-1}, \infty\right)$
 (D) Range of $f(x)$ is $(-\infty, -1] \cup (1, \infty)$
5. Let $f(x)$ be a non constant twice derivable function defined on $\mathbb{R}$ such that $f(2+x) = f(2-x)$ and $f'\left(\frac{1}{2}\right) = 0 = f'(1)$, then minimum number of roots of the equation $f''(x) = 0$ in $(0, 4)$, is
- (A) 4 (B) 3 (C) 2 (D) 5

Space for Rough Work

6. Let $F(n, x) = \sum_{r=1}^n \tan^{-1} \left(\frac{2x}{4 + r(r+1)x^2} \right)$, where $x > 0$ and $n \in \mathbb{N}$ and $\frac{d}{dx} F(n, x) = G(n, x)$, then
- (A) $F(8, 2) = \tan^{-1} \left(\frac{4}{5} \right)$ (B) $\lim_{n \rightarrow \infty} F(n, 2\sqrt{3}) = \frac{\pi}{6}$
- (C) If $G\left(n, \frac{1}{3}\right) < 0$, then $n \geq 36$ (D) $\lim_{n \rightarrow \infty} G(n, 1) = -\frac{2}{5}$
7. If $a + b + c = 4$ and $a^2 + b^2 + c^2 + 3(ab + bc + ac) = 21$ where $a, b, c \in \mathbb{R}$, then which of the following is/are true
- (A) minimum value of $(a+b)(b+c)(a+c)$ is 18
- (B) minimum value of $(a+b)(b+c)(a+c)$ is 24
- (C) maximum value of $(a+b)(b+c)(a+c)$ is $\frac{490}{27}$
- (D) maximum value of $(a+b)(b+c)(a+c)$ is $\frac{471}{27}$
8. Let $f(x) = \begin{cases} x[x] & -4 \leq x < -1 \\ 1-x & -1 \leq x < 1 \\ x^2 - 3x + 2 & 1 \leq x < 2 \\ (x-2)[x] & 2 \leq x \leq 4 \end{cases}$, then (where $[.]$ denotes greatest integer function)
- (A) number of points where $f(x)$ is non-differentiable is 6
- (B) sum of global maximum value and global minimum value of $f(x)$ is $\frac{63}{4}$
- (C) number of points of extrema of $f(x)$ is 3 points
- (D) number of solutions of $f(x) = 4$ is 3

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 9 and 10

Let $f(x) = \frac{1}{2} \left[f(xy) + f\left(\frac{x}{y}\right) \right]$ for $x, y \in \mathbb{R}^+$ such that $f(1) = 0$, $f'(1) = 2$ then

9. $f(x) - f(y)$ is
- (A) $f\left(\frac{y}{x}\right)$ (B) $f\left(\frac{x}{y}\right)$ (C) $f(2x)$ (D) $f(2y)$
10. $f'(3)$ is greater than or equal to
- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Space for Rough Work

Paragraph for Questions 11 and 12

Consider a function $f(x) = ax^3 + x^2 - (2a + 1)x + 2$ (where $a \geq 0$)

11. If the number of value(s) of 'a' for which Rolle's theorem can be applied in the interval $[0, 1]$ is equal to k, then k is less than or equal to
- (A) 0 (B) 1 (C) 2 (D) 4
12. If k_1 and k_2 ($k_1 \neq k_2$) are respectively the local maximum and local minimum values of $f(x)$, then the sum of those roots of the equations $f(x) = k_1$ and $f(x) = k_2$ at which $f'(x) \neq 0$, is
- (A) $\frac{-2}{3a}$ (B) $\frac{-1}{a}$ (C) $\frac{-2}{a}$ (D) $\frac{-1}{3a}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $\int \frac{x^{17}}{(1+x^2)^{10}} dx = \frac{1}{n} \left(\frac{x^2}{1+x^2} \right)^m + C$ where $m, n \in \mathbb{N}$ and C is constant of integration, then value of $\frac{m}{n}$ (in its lowest form) is equal to
2. A circle of radius $2\sqrt{10}$ cm is divided by chord of length 12cm into two parts. If maximum area of rectangle inscribed into smaller circular segment is λ , then $\frac{\lambda}{\sqrt{15}}$ is
3. If the range of the function $y = |f(x)|$, where $f(x) = \frac{195x(x^2+4)}{9x^4+97x^2+144}$ is $\left[0, \frac{a}{b}\right]$ where a and b are relatively prime natural numbers then the value of $a - 4b$ is
4. If $\int \frac{dx}{(\sec x + \operatorname{cosec} x + \tan x + \cot x)^2} = \frac{x}{\alpha} + \frac{\cos\left(x + \frac{\pi}{4}\right)}{\beta} + \frac{\cos 2x}{\gamma} + \delta$, then $|\alpha + \beta + \gamma|$ is

Space for Rough Work

5. If $f(x) = \max\{\min(x, x^2), \max(x, x^2)\}$, where $x \geq 0$ then $\frac{d}{dx} \left((f \circ f)^{-1}(x) \right)$ at $x = 16$ is
6. The value of $\lim_{\theta \rightarrow 0^+} (\sin \theta)^{(\sin \theta - \sin^2 \theta)}$ is equal to
7. If $\frac{d^3 y}{dx^3} = - \left(\frac{dy}{dx} \right)^2 \left[k \left(\frac{d^2 x}{dy^2} \right) \cdot \left(\frac{d^2 y}{dx^2} \right) + \left(\frac{dy}{dx} \right)^2 \left(\frac{d^3 x}{dy^3} \right) \right]$, then k is equal to
8. If $d = (x_2 - x_1)^2 + (-x_2 - 4 + 2\sqrt{x_1 + 2})^2$ where $x_1 > -2$, $x_2 \in \mathbb{R}$, then the minimum value of d is equal to -
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work